

Course Project

[CMP175] - WEB PROGRAMMING

Major: Information Technology

Supervisor: Mr. Trịnh Công Nhựt

Student's name:

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|------------------------------------|----------------|-----------------|
| 1. Full Name : Nguyễn Ngọc Anh Thư | ID: 2380602193 | Class: 23DTHQA2 |
| 2. Full Name : Cao Yến Nhi | ID: 2380601565 | Class: 23DTHQA2 |

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TOPIC 1. ONLINE COURSE WEBSITE

Part 1. Project Introduction (5 Points)

1.1 Problem Statement

- What problem does the website solve?
- Why is this problem important?
- How can the website help users?

1.2 Topic Selection

- Why did the team choose this topic?
- Target users.

1.3 Survey of Existing Websites

Analyze at least TWO similar websites:

For each website:

- Screenshot (Home page)
- Advantages
- Disadvantages
- Features to be implemented

Part 2. System Design (15 Points)

2.1 ERD Diagram

- Required tables: UserAccount, Student, Instructor, Course, Category, Enrollment, Payment, Review
- Recommended additional tables: CourseMaterial, Lesson, CourseStatus

2.2 Usecase Diagram

- Students must draw a Use Case Diagram and explain all user roles.
- User Roles: Guest, Student, Instructor, Admin

2.3 Sample Data

- At least: 10 courses, 10 students, 20 enrollments, 3 instructors, 10 reviews

Part 3. Functions Requirements (70 Points)

Guest (10 points)

- 1/ View Home (Banner, Featured Courses, New Courses, Categories, Navigation Menu, Login/Register Links)
- 2/ View Course List (Course Image, Course Name, Instructor, Category, Price, Search, Filter, Pagination)
- 3/ View Course Detail (Course Information, Description, Instructor Information, Duration, Price, Reviews, Enroll Button)
- 4/ Register (Full Name, Email, Password, Confirm Password, Validation)
- 5/ Login (Email, Password, Remember Me, Forgot Password)

Student (15 points)

- 1/ My Profile (View Profile, Update Profile, Change Password, Upload Avatar)
- 2/ Enroll Course (Login Required, Course Selection, Enrollment Information, Enrollment Confirmation)
- 3/ My Courses (Enrolled Courses, Enrollment Date, Learning Status, Course Details)
- 4/ Payment (Payment Information, Amount, Payment Date, Payment Status)
- 5/ View Course Materials (Lessons, Documents, Video Links, Download Materials)
- 6/ Review Course (Rating, Comment, Review Date, Course Feedback)

Instructor (15 points)

- 1/ Instructor Dashboard (Total Courses, Total Students, Average Rating, Recent Activities)
- 2/ My Teaching Courses (Course List, Course Status, Enrollment Count, Course Information)
- 3/ Manage Courses (Create Course, Edit Course, Update Course Information, Change Course Status)
- 4/ Manage Lessons (Add Lesson, Edit Lesson, Delete Lesson, Upload Learning Materials)
- 5/ Manage Course Materials (PDF Files, Video Links, Documents, Download Resources)
- 6/ View Enrolled Students (Student List, Enrollment Date, Payment Status, Learning Progress)
- 7/ View Course Reviews (Student Reviews, Ratings, Comments, Review Statistics)
- 8/ Course Statistics (Total Students, Total Enrollments, Average Rating, Most Popular Course)

Admin (15 points)

- 1/ Dashboard (Total Courses, Total Students, Total Instructors, Total Enrollments, Total Revenue)
- 2/ Manage Courses (Add Course, Edit Course, Delete Course, Assign Instructor, Course Status)
- 3/ Manage Categories (Add Category, Edit Category, Delete Category)
- 4/ Manage Instructors (Add Instructor, Edit Instructor, Activate/Deactivate Account)
- 5/ Manage Students (View Students, Edit Student Information, Activate/Deactivate Account)
- 6/ Manage Enrollments (View Enrollment List, Update Enrollment Status, Cancel Enrollment)
- 7/ Manage Payments (View Payments, Confirm Payment, Update Payment Status)
- 8/ Manage Reviews (View Reviews, Delete Inappropriate Reviews, Review Statistics)

Additional Features (10 points)

- a) Email Notification (Account Verification, Password Reset, Enrollment Confirmation, Booking Confirmation, Payment Confirmation)
- b) Online Payment (PayPal, VNPay, MoMo, Stripe, Payment Confirmation, Transaction History)
- c) Deployment (Publish Website, Deploy to Hosting, Deploy to Cloud Platform, Public Access URL)
- d) Dashboard Enhancement (Charts, Revenue Statistics, User Statistics, Top Data Reports)
- e) File Upload (Profile Image, Product Image, Course Materials, Documents)
- f) Security Enhancement (Role-based Authorization, Password Encryption, Account Lockout)

UI/UX (5 Points)

- a) Responsive
- b) Validation
- c) Upload Image
- d) Error Handling

Part 4. Project Understanding Test (15 points)

- a) Question 1. Database Understanding
- b) Question 2. Function Understanding
- c) Question 3. Basic Code Modification

Important Notes

1. Understand the Requirements

- The requirements in this document represent the customer's expectations.
- Your task is to analyze the problem and develop a complete solution that satisfies the required functions.
- Different teams may have different designs and implementations.

2. Database Design

The database tables provided are only suggestions.

Each team is responsible for designing:

- Tables and attributes
- Primary Keys (PK)
- Foreign Keys (FK)
- Relationships between tables

Your database must support all required functions of the system.

3. Full-Stack Development

Each team must develop both:

- Frontend
- Backend

Using existing projects, copied source code, or reused backend systems is strictly prohibited.

() Any detected plagiarism or project reuse will result in a score of 0.*

4. Report Requirements

For every required function:

- Provide screenshots of the result.
- Place screenshots in the correct section of the report.
- Clearly demonstrate the implemented functionality.

5. Project Schedule

Week 14

- Complete the website.
- Demonstrate all required functions.
- Project evaluation.

Week 15

- Submit Final Report.
- Submit Source Code.
- Submit Database Script.
- Complete the Project Understanding Test.

6. Team Responsibility

Every team member should understand:

- Database design
- System workflow
- Main functions
- Source code structure

Individual understanding will be considered during the final evaluation.

7. Use of AI Tools

AI tools may be used for learning and development support.

However:

- You must understand the generated code.
- You must be able to explain your implementation.
- You must be able to modify the system independently.

8. Academic Integrity

The following actions are prohibited:

- Copying code from other teams.
- Reusing projects from previous semesters.
- Purchasing completed projects.

() Violations may result in a score of 0.*

9. Evaluation Criteria

The final score is based on:

- Project quality
- Database design
- Functional implementation
- UI/UX quality
- Project understanding
- Ability to explain and modify the system

Final Project Topic Registration

- Each team consists of **2-3 students**.
- Each team selects **ONE topic** from the list below:

Topic 1: Online Course Website

Topic 2: Language Center Website

Topic 3: Job Portal Website

Registration Format

Team X (Full Name 1; Full Name 2; Full Name 3)

Topic : Topic Name

Example

Team 01 (Nguyen Van A; Tran Thi B; Le Thi C)

Topic 1: Online Course Website

Notes

- Maximum **4 teams per topic**.
- First come, first served.
- Please comment directly under this announcement.
- A topic is officially assigned only after instructor confirmation.

Topic 1: Online Course Website

Topic 2: Language Center Website

Topic 3: Clinic Website

Topic 1: Online Course Website

Topic 2: Language Center Website

Topic 3: Library Management Website

Topic 4: Hotel Website

Topic 2: Gym Management Website

